Emerging Blockchain Systems: **Polkadot and How it Works: Blockchain of Blockchains (Par Chains)**

In a nutshell

- Polkadot is a decentralized, scalable, and interoperable blockchain network.
- It is built on a central Relay Chain and connected Parachains.
- Polkadot's architecture allows for the development of custom blockchains while maintaining security and interoperability.
- DOT token is used for governance, staking, and network security.
- Upgrades can be applied without the need to fork
- It adopts an autonomous democratic governance approach to vote on big decisions and changes
- Adopts a Nominated Proof of Stake Approch (Npos)

What is Polkadot

Polkadot is a blockchain platform that enables different blockchains to communicate and share information with each other, creating an interconnected network. It is actually often described as a “blockchain of blockchains.”

- Polkadot is a decentralized network that aims to connect and secure different blockchains. It provides a framework for developers to build their custom blockchains while benefiting from the security and interoperability of the underlying network. As a result is often described as a “blockchain of blockchains”.
- As of 2026, Polkadot is highly relevant in the blockchain ecosystem due to its modular framework.

The Story

- In 2017, the Web3 Foundation, co-founded by Wood, held an initial coin offering (ICO) that raised approximately $144 million to fund the development of Polkadot's project.
- Despite a setback when $90 million worth of funds were frozen due to a Parity wallet vulnerability, the project continued with the remaining resources and later held another funding round. Polkadot had its initial block, the genesis block, in 2020.
- Polkadot mainnet version is referred to as Polkadotv1 and was focused on the main architecture of the project. Polkadot 2.0 results from many upgrades and a more mature network stage.
- Polkadot allows a wide range of blockchains to interoperate because it works on a blockchain protocol that facilitates the cross-chain transfer of any data or asset type, not just tokens. This allows for the creation of specialized blockchains (parachains) that connect to a universal network (the Relay Chain).

Main Components of Polkadot

Polkadot works with three specific components that make it unique: the relay chain, parachains, and bridges.

- Relay Chain
  - This is Polkadot's main blockchain that coordinates the system, handles interoperability, and provides shared security.
  - Polkadot uses a consensus mechanism called nominated proof-of-stake (NPoS)
  - NPoS ensures the integrity of the blockchain. It also provides a decentralized governance system and validates transactions.
- Parachains
  - These are independent blockchains connected to the relay chain, and each of them can have their own tokens.
  - Polkadot's parachains provide customizability and scalability.
  - For example, a parachain can specialize in identity verification while another one specializes in real estate transactions.
- Bridges
  - Bridges are special parachains that allow Polkadot to connect and communicate with other blockchains like Ethereum.
  - As a result, they facilitate interoperability and expand the network's capabilities, like cross-chain communication and asset transfer.

How it Works?

- According to its official website, Polkadot is “a network governed by a rebelliously innovative community.” Its native token DOT plays a crucial role in the development and implementation of the network, and it has multiple functions:
- **Governance:** DOT holders can participate in governance decisions, including determining network fees, auction dynamics, the addition or removal of parachains, and changes to the protocol.
- **Staking:** Involves locking up DOT tokens to support network operations and integrity. Validators and nominators are selected based on the amount of DOT they stake. Stakes also help secure the network and earn rewards.
- **Bonding and parachain auctions:** In Polkadot, new parachains join the network through a process called parachain auctions. To secure a slot, projects compete in these auctions by bonding (locking up) DOT tokens. The project that commits the most DOT during the auction wins the slot, granting it the right to operate as a parachain for a fixed period, usually up to two years.

How it Works?

- **Transaction processing and finality**: Polkadot uses advanced mechanisms to ensure quick transaction processing and finality, helping to improve the security of the Relay Chain.
- **Scalability and performance advantages:** Polkadot achieves scalability by enabling parallel transaction processing across multiple parachains. Each parachain operates independently and can process transactions simultaneously, rather than all transactions being handled sequentially on a single chain (as is the case with traditional blockchains like Bitcoin or Ethereum).

Proof of History

Polkadot's design offers several benefits that address some of the primary challenges faced by earlier blockchain technologies.

- **Interoperability:** Enables different blockchains to interact and share information.
- **Scalability:** Multiple parachains process transactions in parallel, significantly increasing throughput.
- **Security:** Shared security model where all parachains benefit from the robust security of the Relay Chain.
- **Customizability:** Developers can create customized blockchains that are tailored to specific needs.
- **Ecosystem development:** Encourages a collaborative environment for developers, fostering innovation.

Polkadot New Features and Advantages

Sharded multichain network

- Polkadot has a distinct advantage compared to traditional blockchains in that it uses a sharded multichain approach. This allows many transactions to be processed on multiple chains simultaneously. This makes it much more scalable, opening the door to wider use cases and development. Parachains, which are interconnected blockchains, are run in parallel within the Polkadot system.

Customizable blockchain architecture

- Polkadot knows that a single set of parameters is unsuitable for all blockchain designs. Its adaptability lets each blockchain be customized to suit the purpose for which it's intended. This versatility results in greater efficiency, security, and better services, as blockchains can eliminate redundant code and concentrate on their primary functions.

Interoperability and cross-chain communication

- Polkadot allows networks and applications to cooperate in an effortless manner, similar to how apps on a mobile phone interact with each other. This is a major change from the isolated nature of earlier networks. This opens up the possibility of developing cutting-edge services and transmitting information between blockchains.

Polkadot New Features & Advantages

Democratic network governance

- Polkadot's networks allow communities to administer their networks in a clear and democratic manner autonomously. This flexibility also applies to their governance structures, which can be modified or improved as needed.

Forkless upgrades

- Polkadot removes the need for hard forks, which can cause divisions among users. Hard forks also require a lot of effort to address a widespread issue in blockchain technology — the requirement for upgrades. This allows blockchains within the Polkadot network to advance and adjust more fluidly as technology progresses.
- Polkadot's scalable, customizable, interoperable architecture, combined with its democratic governance and efficient upgrade abilities, puts it ahead of the pack in the blockchain industry. This pioneering combination makes Polkadot a leader in the advancement of blockchain technology.

Nominated Proof of Stake (Npos)

- Polkadot utilizes a nominated proof-of-stake (NPoS) consensus mechanism involving validators and nominators.
- Validators secure the Relay Chain by staking DOT, validating transactions on the parachains, participating in consensus with other validators and producing blocks on the Relay Chain.
- Nominators, on the other hand, contribute to the network's security by staking their DOT to back trustworthy validators. This system enhances the network's security while enabling cross-chain communication and interoperability.
- Polkadot also includes two specialized roles: Collators, who manage and submit valid parachain transactions to Relay Chain validators, and Fishermen, who monitor and report network misconduct.
- These roles require more technical skill than nominators but less commitment than full validators, playing a vital role in maintaining the network.